UNIVERSITY OF SOUTH CAROLINA UPSTATE

Master of Science in Computer Science

Program Feasibility Study and Market Analysis for a Proposed Online Graduate Degree

PREPARED FOR	PROGRAM FORMAT	REPORT DATE	PROPOSED LAUNCH
USC Upstate Academic Affairs	**Online / Hybrid**	**June 2026**	**Fall 2027**

Executive Summary

This feasibility study examines the case for USC Upstate to offer a fully online Master of Science in Computer Science (MSCS). The analysis covers national and regional labor market conditions, a review of comparable graduate programs, a proposed curriculum structure, five-year financial projections, institutional risks, and a recommended implementation path.

Computer Science graduate education is among the highest-demand segments in American higher education. The Bureau of Labor Statistics projects that software developer, quality assurance analyst, and related computing occupations will grow 25 percent through 2032.[1] That growth is not evenly distributed: roles in artificial intelligence, machine learning, cloud computing, and data engineering are growing at rates two to three times the broader category average.[2] South Carolina's expanding technology sector is feeling that pressure directly, and the region around Spartanburg and Greenville has limited graduate CS education options for working professionals.

25%	$132K	4.2M	$2.6M
10-YEAR JOB GROWTH FOR CS ROLES (BLS)	MEDIAN MS COMPUTER SCIENCE SALARY	COMPUTING JOB OPENINGS PROJECTED BY 2030	PROJECTED CUMULATIVE NET REVENUE (YR 1–5)

> **Recommendation: Proceed.** A USC Upstate MS in Computer Science is financially viable, educationally sound, and regionally needed. The program can reach break-even with 30 active students and is modeled to produce positive net revenue by Year 2. Timing matters — the Upstate tech corridor is growing rapidly, and early program establishment will be substantially harder to replicate once a competitor enters the space.

Scope of This Study

The analysis draws on national labor market data from the Bureau of Labor Statistics, Computing Research Association workforce reports, South Carolina labor market statistics, a competitive review of online CS graduate programs at comparable institutions, and internal cost modeling based on USC Upstate's current operational infrastructure and faculty profile. All enrollment and revenue figures represent planning estimates and should be reviewed annually against actual performance.

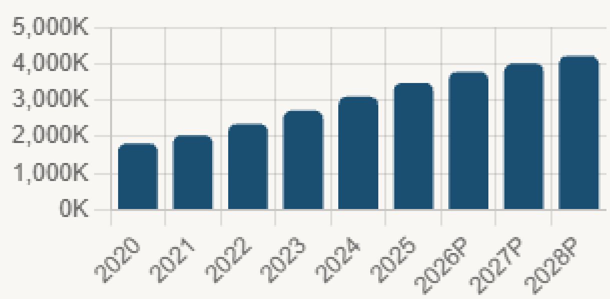
Market Analysis

National Computing Workforce Demand

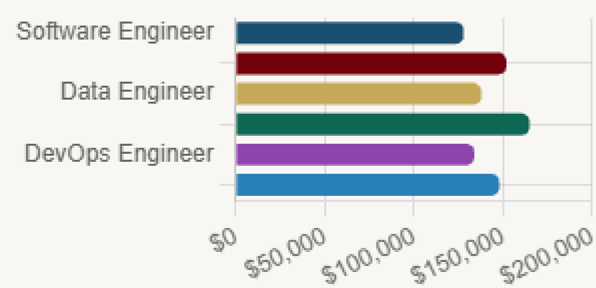
Computing is one of the fastest-growing occupational categories in the United States, and graduate-level credentials are increasingly required for senior individual contributor and leadership roles. The Computing Research Association's 2025 Taulbee Survey found that the number of CS PhD graduates remained flat while industry demand for advanced degree holders continued to rise — a supply-demand gap that is now translating into strong demand for professionally oriented master's programs.[3]

Areas of particular growth include machine learning and AI engineering, cloud infrastructure and DevOps, full-stack software development, data science and analytics engineering, and distributed systems. These are not niche specializations anymore — they are mainstream requirements across manufacturing, healthcare, finance, and logistics.[4]

U.S. COMPUTING JOB OPENINGS — 2020 TO 2028 (THOUSANDS, PROJECTED)



MEDIAN SALARIES BY COMPUTER SCIENCE SPECIALIZATION



South Carolina and Upstate Regional Demand

South Carolina added more than 12,000 technology sector jobs between 2020 and 2025, a 31 percent increase that outpaced the national average.[5] The Upstate region accounts for a disproportionate share of that growth, driven by the presence of international manufacturers who have built sophisticated software and data operations alongside their physical plants — BMW, Michelin, Milliken, and others now employ substantial in-house software development and data analytics teams locally.

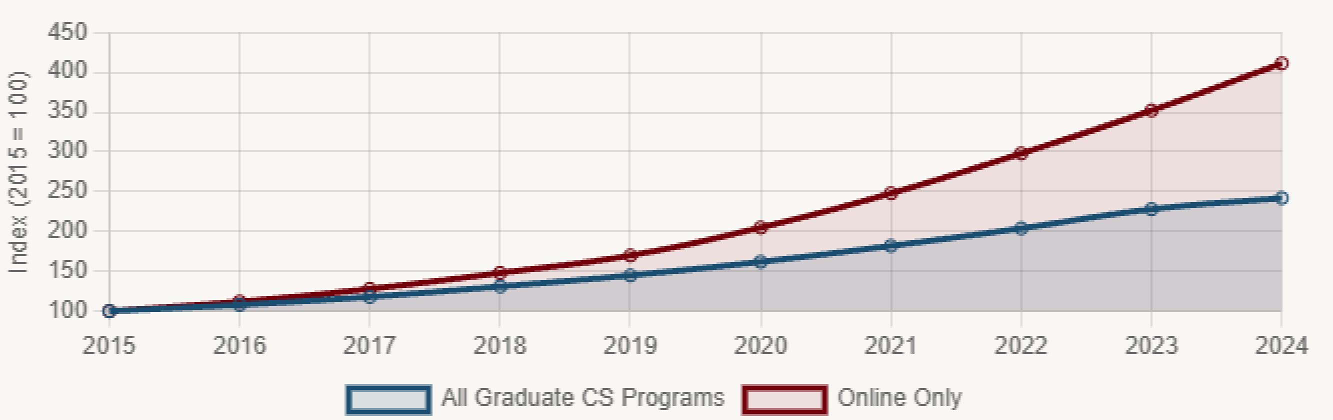
South Carolina's Department of Employment and Workforce recorded 2,214 active computing and software job postings in Q1 2026, with the largest clusters in Greenville, Spartanburg, and the I-85 corridor. Average advertised salaries for roles requiring a master's degree were $138,200 — roughly $22,000 higher than the same roles open to bachelor's-degree applicants.[6]

Sector	Active CS Openings (Q1 2026)	Year-over-Year Change	Avg. Salary (MS Required)
Software and Technology	618	**+24%**	$138,400
Manufacturing and Advanced Mfg.	487	**+31%**	$128,700
Healthcare and Life Sciences	392	**+28%**	$134,500
Financial Services and FinTech	341	**+18%**	$148,200
Logistics and Supply Chain	218	**+36%**	$118,900
Government and Defense	158	**+21%**	$125,000
Total	**2,214**	**+27%**	**$132,300 avg.**

Graduate CS Enrollment Trends

Online master's programs in Computer Science have shown consistent enrollment growth over the past decade. The National Center for Education Statistics reported a 42 percent increase in graduate CS enrollment between 2015 and 2024, with online programs accounting for the majority of that growth since 2020.[7] That trend reflects both the proven quality of online delivery for technical subjects and the strong preference of working professionals for flexible scheduling.

GRADUATE CS ENROLLMENT GROWTH — NATIONAL ONLINE PROGRAMS (INDEX, 2015 = 100)



Competitive Landscape

The national online MSCS market is anchored by a small number of very large, very well-known programs. However, the regional picture — specifically for South Carolina working professionals who want a university name they recognize and a price point that makes sense for a self-funded or employer-sponsored student — is considerably more open than the national view suggests.

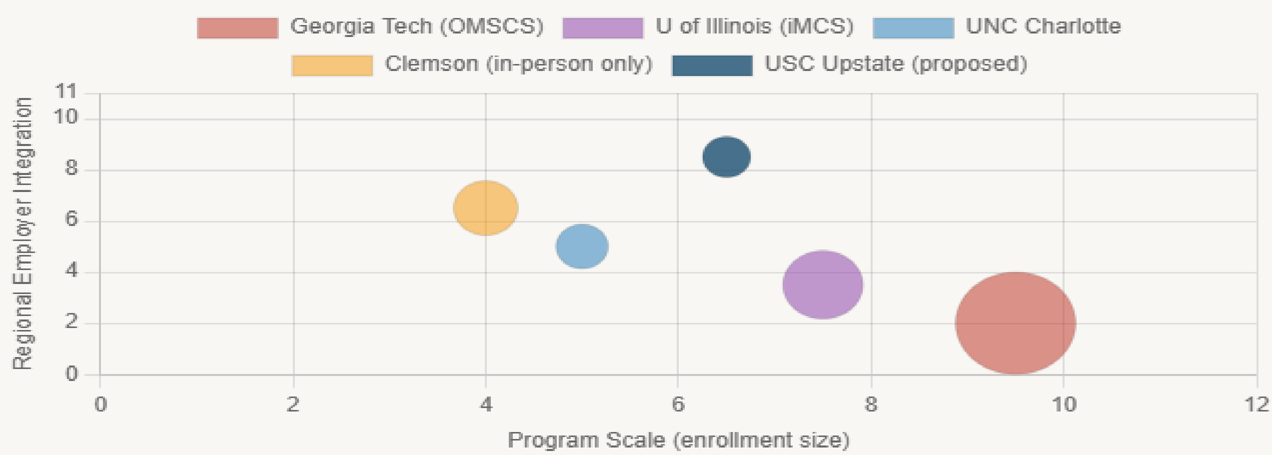
Institution	Program	Format	Credits	Tuition / Credit	Notes
Georgia Tech (OMSCS)	MS Computer Science	Online	30	$182	Dominant brand; exceptionally low price; very high volume (10,000+ enrolled)
University of Illinois	MS Computer Science (iMCS)	Online	32	$636	Strong brand; Coursera partnership; primarily asynchronous
Clemson University	MS Computer Science	In-person only	30	$642 (in-state)	No online option; requires campus presence — significant barrier for working adults
UNC Charlotte	MS Computer Science	Hybrid	30	$688	North Carolina market; minimal Upstate SC brand relevance
Purdue Global	MS Information Technology	Online	36	$720	For-profit; broad but not CS-specific depth
Winthrop University	No MS CS	—	—	—	**Direct regional peer with no competing offering**
USC Upstate (Proposed)	**MS Computer Science**	**Online**	**33**	**$725**	**Only public online MSCS in Upstate SC**

The Georgia Tech Question

The OMSCS program at Georgia Tech — with more than 10,000 enrolled students and a tuition rate of $182 per credit — is the elephant in the room for every institution considering an online MSCS.[8] It is genuinely excellent and genuinely inexpensive. However, the OMSCS experience is designed for a specific type of student: highly selfdirected, comfortable with a largely asynchronous format, willing to wait in queues for popular courses, and unconcerned with personal advising or regional employer connections.

USC Upstate's comparative advantage lies precisely in the areas where OMSCS is structurally thin: smaller cohorts, personal faculty relationships, dedicated career advising, integration with Upstate SC employer networks, and a degree from an institution that local hiring managers know and trust. These are not small differences for the working professional market.

POSITIONING MAP — PROGRAM SIZE VS. REGIONAL EMPLOYER INTEGRATION



Proposed Program Structure

Program Overview

Degree	Master of Science in Computer Science (MSCS)
Total Credit Hours	33 (11 courses at 3 credits each)
Delivery Format	Fully online; optional on-campus intensive sessions each semester
Expected Duration	18–24 months full-time; up to 36 months part-time
Admission Requirements	Bachelor's degree in Computer Science, Mathematics, or related STEM field; minimum 3.0 GPA; statement of purpose; programming proficiency demonstrated by transcript or assessment
Concentrations	Artificial Intelligence and Machine Learning; Software Engineering and Cloud Systems; Data Science and Analytics
Thesis Option	Available for students intending doctoral study; replaces two electives and capstone with six-credit thesis

Curriculum

The proposed curriculum reflects ACM Computing Curricula 2023 guidelines for graduate CS programs[9] and is designed to serve three audiences simultaneously: working professionals seeking advancement in their current technical track, career changers from adjacent STEM fields, and students considering doctoral study. The core establishes theoretical and systems foundations; the electives allow meaningful specialization within one concentration area.

Course Number	Course Title	Credits	Type
CSCI 700	Analysis of Algorithms	3	Core
CSCI 710	Advanced Software Engineering	3	Core
CSCI 720	Machine Learning and Statistical Modeling	3	Core
CSCI 730	Distributed Systems and Cloud Computing	3	Core
CSCI 740	Database Systems and Data Engineering	3	Core

Course Number	Course Title	Credits	Type
CSCI 750	Artificial Intelligence and Reasoning	3	Core
CSCI 760	Human-Computer Interaction and UX	3	Core
CSCI 770	Deep Learning and Neural Networks	3	Elective
CSCI 780	Natural Language Processing	3	Elective
CSCI 790	DevOps, MLOps, and Platform Engineering	3	Elective
CSCI 799	Capstone Project or Thesis	3	Capstone

Industry and Certification Alignment

Graduate technology credentials command strong premiums in the job market, and coursework in this program prepares students for widely recognized vendor certifications that further increase their market value.[10]

Certification	Issuing Organization	Aligned Courses	Avg. Salary Premium
AWS Solutions Architect — Professional	Amazon Web Services	CSCI 730, 790	+$18,000
Google Professional Machine Learning Engineer	Google Cloud	CSCI 720, 770	+$22,000
Microsoft Azure AI Engineer Associate	Microsoft	CSCI 750, 780	+$17,000
Databricks Certified Data Engineer	Databricks	CSCI 740	+$15,000
PMI Agile Certified Practitioner (PMI-ACP)	PMI	CSCI 710	+$11,000

Financial Projections

Planning Assumptions

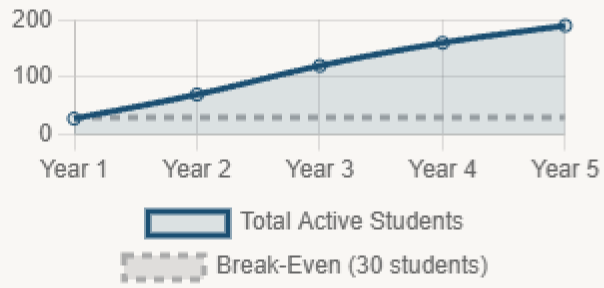
Tuition rate	$725 per credit hour (online rate, in-state and out-of-state)
Total credits per student	33 hours
Revenue per student (full program)	$23,925
Faculty cost per course section	$9,000–$13,000 (adjunct); tenure-track salary prorated by load
Online platform	USC Blackboard (existing infrastructure; incremental cost approximately $15,000 per year)
Year 1 marketing budget	$130,000
Subsequent years marketing	$85,000 per year
Break-even enrollment threshold	30 active students

Five-Year Enrollment and Revenue Forecast

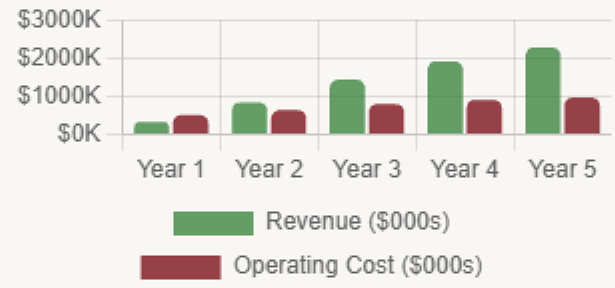
The projection below assumes a conservative Year 1 cohort of 28 students, growing to 190 active students by Year 5. Peer program data from comparable regional online MSCS launches support these assumptions as achievable without exceptional marketing spend.[11]

Year	New Students	Total Active	Gross Revenue	Operating Cost	Net
Year 1 (FY 2028)	28	28	$335,950	$520,000	**–$184,050**
Year 2 (FY 2029)	45	70	$839,750	$640,000	**+$199,750**
Year 3 (FY 2030)	65	120	$1,439,100	$800,000	**+$639,100**
Year 4 (FY 2031)	80	160	$1,919,000	$910,000	**+$1,009,000**
Year 5 (FY 2032)	95	190	$2,278,000	$975,000	**+$1,303,000**

ENROLLMENT GROWTH BY YEAR



REVENUE VS. OPERATING COST (FIVE-YEAR)



Start-Up Cost Summary

Cost Category	Year 1	Year 2	Year 3 and Beyond
Curriculum development (11 courses)	$160,000	$35,000	$20,000
Faculty (2 tenure-track lines + adjuncts)	$185,000	$195,000	$210,000
Technology and cloud lab environment	$55,000	$22,000	$15,000
Marketing and student recruitment	$130,000	$85,000	$85,000
Accreditation and ABET consultation	$40,000	$18,000	$10,000
Administrative support staffing	$45,000	$50,000	$55,000
Total	**$615,000**	**$405,000**	**Approximately $395,000**

SWOT Analysis

This analysis reflects institutional conditions as of June 2026 and draws on internal program data, peer institution benchmarks, and SACSCOC substantive change requirements.[12]

STRENGTHS

- Active BS in Computer Science program provides an established faculty base and curricular infrastructure
- USC system institutional credibility recognized by South Carolina employers
- SACS-COC regional accreditation in good standing
- Online tuition rate competitive against all regional alternatives except Georgia Tech
- Existing Blackboard LMS and IT infrastructure reduce platform start-up cost
- Growing undergraduate CS enrollment increases graduate feeder pool over time

WEAKNESSES

- No current graduate CS program — building from zero enrollment base
- Faculty research profile in high-demand areas (ML, AI) still developing
- No dedicated graduate computing research lab or funded research grants at present
- National brand awareness lower than Georgia Tech, Clemson, or UofSC Columbia
- Start-up investment of $615,000 in Year 1 requires advance funding commitment

OPPORTUNITIES

- No competing public online MSCS in Upstate South Carolina
- Clemson's in-person-only format leaves Upstate working professionals underserved
- NSF CISE Research Initiation Initiative grants available for new CS faculty
- Employer tuition benefit pipeline through BMW, Michelin, and Milliken
- Greenville–Spartanburg tech corridor growth projected to continue through 2030
- State GEAR UP and LIFE scholarship extensions to graduate level under discussion

THREATS

- Georgia Tech OMSCS at $182 per credit is structurally difficult to match on price
- University of Illinois iMCS leverages Coursera's massive marketing reach
- Faculty recruitment is extremely competitive given industry salary levels for CS
- AI-driven changes in software development may reshape required skills faster than curricula can adapt
- SACSCOC substantive change review process adds 12–18 months minimum to launch

Implementation Timeline

Q3 2026 — APPROVAL PHASE

Governance Approval and SACSCOC Notification

Present feasibility study to Faculty Senate and Academic Affairs. Submit SACSCOC Substantive Change notification for new online graduate program. Begin faculty search planning.

Q4 2026 — DEVELOPMENT PHASE

Curriculum Build-Out and Faculty Hiring

Post two tenure-track faculty searches in AI/ML and Software Engineering. Contract instructional designers. Develop 11 course shells in Blackboard. Finalize concentration structure.

Q1 2027 — INFRASTRUCTURE PHASE

Cloud Lab Environment and Industry Advisory Board

Launch cloud-based computing lab (AWS Educate or Azure for Academia). Form Industry Advisory Board with Upstate SC technology employer representatives. Apply for NSF CISE RII grant.

Q2 2027 — RECRUITMENT PHASE

Marketing Campaign and Application Review

Launch targeted digital campaign for Upstate SC working professionals. Open admissions. Activate employer tuition partnership outreach with top 10 regional technology employers.

FALL 2027 — PROGRAM LAUNCH

First Cohort Begins

Target 28 students in inaugural cohort. Begin tracking retention, satisfaction, and time-to-completion metrics. Initiate ABET program evaluation consultation.

FY 2028–2029 — SUSTAINABILITY PHASE

Scale Enrollment and Achieve Program Self-Sufficiency

Program reaches financial sustainability at approximately 70 active students. Pursue ABET accreditation. Explore thesis track expansion and research grant applications.

Recommendations and Next Steps

Overall Recommendation: Proceed with program development. The MSCS program addresses a genuine and growing regional workforce need, aligns with institutional strategic goals for graduate program expansion, and presents a financially sustainable path that recovers start-up costs within two years of launch. The competitive window for first-mover advantage in the Upstate SC online graduate CS market is open now and will close as the region's technology sector growth attracts competitor attention.

Priority Actions

Priority	Action	Owner	Target Date
1 — Critical	Secure Board of Trustees approval for program development	Provost Office	Q3 2026
2 — Critical	File SACSCOC Substantive Change notification for new online graduate program	Accreditation Officer	Q3 2026
3 — Critical	Commit start-up funding ($615,000 Year 1) from grants and institutional reserves	CFO and Development Office	Q3–Q4 2026
4 — Critical	Hire MSCS Program Director and initiate two faculty searches	Dean, College of Science and Engineering Technology	Q4 2026
5 — High	Form Industry Advisory Board with six to eight Upstate SC technology employer representatives	Program Director	Q4 2026
6 — High	Apply for NSF CISE Research Initiation Initiative grant for new faculty	Program Director and Sponsored Research	Q1 2027
7 — High	Launch digital recruitment campaign targeting Upstate SC working professionals	Marketing and Communications	Q2 2027
8 — Medium	Establish employer tuition benefit agreements with top 10 regional technology employers	Business Development	Q2 2027
9 — Medium	Begin ABET program evaluation consultation for future accreditation	Program Director	Fall 2027

Risk Mitigation Summary

Risk	Likelihood	Impact	Mitigation Strategy
Below-target Year 1 enrollment	Medium	High	Pre-launch employer cohort commitment from two or three partners; adjust marketing allocation based on application velocity
Faculty recruitment in AI and ML	High	High	Recruit industry practitioners for elective courses; offer competitive market salary; use endowed chair funding if available
Georgia Tech OMSCS competition	High	Medium	Target students who value smaller cohorts, regional employer connections, and personal advising — segments where OMSCS structurally underdelivers
Curriculum keeping pace with AI advances	Medium	Medium	Annual curriculum review process built into program governance; Industry Advisory Board engagement on emerging skill needs
SACSCOC approval delay	Low	High	File substantive change early; schedule pre-consultation with SACSCOC staff before formal submission

Closing Statement

The case for an online MSCS at USC Upstate comes down to a straightforward observation: the Upstate South Carolina region is growing into a genuine technology hub, and its workforce development infrastructure has not kept pace. Skilled professionals in Spartanburg and Greenville who want a master's degree in Computer Science today must either commute to Clemson — if they can arrange a schedule that allows it — or enroll in a large national online program that has no particular stake in their local job market.

USC Upstate can offer something different: a graduate degree from a university with real roots in the community, taught by faculty who understand the regional employers, priced within reach of a working professional's budget. That combination is worth building. The five-year financial model shows it pays for itself. The market data shows the demand is there. The question is whether the institution moves before someone else does.

University of South Carolina Upstate

800 University Way, Spartanburg, SC 29303
upstate.edu

Prepared by: Office of Academic Planning and Innovation — June 2026

This document is intended for internal institutional planning. All enrollment and revenue projections are estimates based on available market data and peer program benchmarks.

References

1. U.S. Bureau of Labor Statistics. (2024). *Occupational Outlook Handbook: Software Developers, Quality Assurance Analysts, and Testers.* Washington, DC: U.S. Department of Labor. Retrieved from https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm
2. McKinsey Global Institute. (2025). *The State of AI in 2025: Workforce and Skills Implications.* New York: McKinsey and Company. Retrieved from https://www.mckinsey.com/ai-2025
3. Computing Research Association. (2025). *Taulbee Survey: Production of PhDs, Employment of PhD Graduates, and Faculty Salaries in North American Computer Science and Computer Engineering Programs.* Washington, DC: CRA. Retrieved from https://cra.org/resources/taulbee-survey
4. LinkedIn Economic Graph. (2025). *Jobs on the Rise: United States — 2025 Edition.* Sunnyvale, CA: LinkedIn Corporation. Retrieved from https://economicgraph.linkedin.com
5. South Carolina Department of Commerce, Research Division. (2025). *South Carolina Technology Sector Employment Report: 2020–2025.* Columbia, SC: SCDOC. Retrieved from https://sccommerce.com/research
6. South Carolina Department of Employment and Workforce. (2026). *Q1 2026 Occupational Demand Survey — Computing and Software Occupations.* Columbia, SC: SCDEW Labor Market Information. Retrieved from https://www.scdew.state.sc.us/labor-market
7. National Center for Education Statistics. (2025). *Digest of Education Statistics 2024 — Table 323.10: Degrees in Computer and Information Sciences.* Washington, DC: U.S. Department of Education. Retrieved from https://nces.ed.gov/programs/digest
8. Georgia Institute of Technology. (2026). *Online Master of Science in Computer Science (OMSCS) — Program Statistics.* Atlanta, GA: Georgia Tech College of Computing. Retrieved from https://omscs.gatech.edu
9. ACM / IEEE-CS / AAAI Joint Task Force. (2023). *Computing Curricula 2023: Paradigms for Global Computing Education.* New York: Association for Computing Machinery. Retrieved from https://cc2023.acm.org
10. Global Knowledge. (2025). *IT Skills and Salary Report 2025.* Cary, NC: Global Knowledge Training. Retrieved from https://www.globalknowledge.com
11. EDUVENTURES Research. (2024). *Online Graduate Program Enrollment Benchmarks: Regional Public Universities.* Boston, MA: Eduventures. Retrieved from https://www.eduventures.com/research
12. Southern Association of Colleges and Schools Commission on Colleges. (2024). *SACSCOC Policy on Substantive Change.* Decatur, GA: SACSCOC. Retrieved from https://sacscoc.org